INCH-POUND

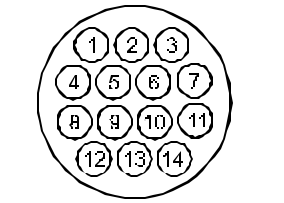
MIL-DTL-16377/77 (SH)
27 September 2004

DETAIL SPECIFICATION SHEET

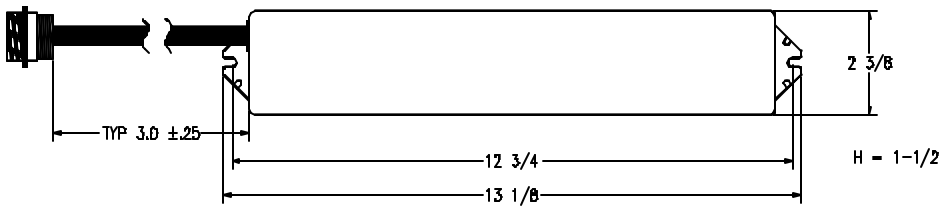
FIXTURES, LIGHTING; EMERGENCY BALLAST, FOR ONE 20-WATT FLUORESCENT LAMP

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this document and the latest issue of MIL-DTL-16377.



CONTACT ARRANGEMENT



CONTACT ARRANGEMENT	WIRE COLOR
1	Red w/ White Stripe
2	White w/ Red Stripe
3	Red w/ White Stripe
4	No Connection
5	White w/ Red Stripe
6	Blue w/ White Stripe
7	No Connection
8	No Connection
9	Blue
10	White
11	No Connection
12	Green
13	Red
14	Yellow

LABEL TYPE I

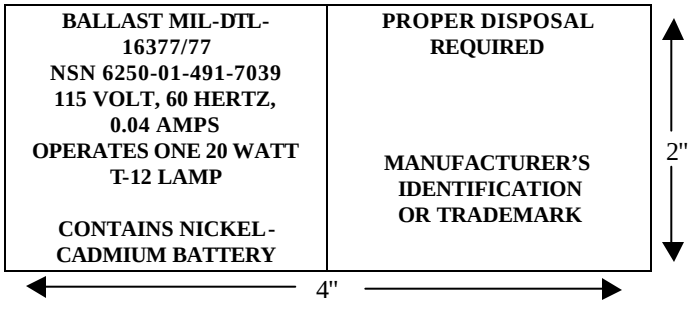


FIGURE 1. Emergency ballast with wiring harness.

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REQUIREMENTS:

DESIGN:

Electrical rating: 20 watts, 120 volts, 60 hertz.

Dimension and configurations: See Figure 1.

Assembly: The emergency ballast consists of a high temperature, rechargeable NI-CAD battery; battery charger; A.C. power sensing circuit; solid state inverter; low voltage battery cutoff, wiring harness and enclosure. UL approved ballast may be furnished if it conforms to the dimensions and requirements specified.

Weight: 4 lb. maximum

Performance characteristics:

Input voltage 115 VAC ± 10%	
Emergency operating time	5.0 hrs. (min)
Full recharge time	20 hrs.
Hi-temp NiCad battery	24 Wh @ 55 °C
Battery rated voltage	6.0 volts
Battery discharge current	0.8 amps
Battery charge current	0.2 amps
Charging component of AC line	0.04 amps
Charging power	4.0 watts
System power factor	0.9 (min)
System harmonics	< 3%
Low voltage cutoff	4.0 volts
Lumen output @ 0 hrs.	240 lumens
Lumen output @ 5 hrs.	140 lumens

Marking: Ballast shall conform to marking requirements and color coding of leads as specified in the applicable paragraph of American National Standards Institute (ANSI) standard for “fluorescent lamp ballast’s, No. C82.1”.

Label: The labels shall be in accordance with type I except as specified herein. (see Figure 1)

Standard Parts: The standard parts and their respective part numbers shown in Table I shall be used in the construction of the fixtures specified herein.

TABLE I. Standard parts.

Part Number	Description	Specification/Material	Remarks
M16377/77-001	Emergency ballast with wiring harness	AMP part No. 206043 or equal	Mates to AMP part No. 206043-1

EXAMINATION AND TESTS: Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:

Complete product: See Table II.

Conformance:

Comparison inspection: Same as the first article inspection.

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TABLE II. First article and comparison inspection for complete product.

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Performance (a) Power output (b) Lumen output (c) System power factor (d) Operating time	1/	2/	3/
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Battery discharge test 4/	-	-	

1/ The performance of the ballast shall be in accordance with the requirements listed in the applicable paragraph of ANSI standard for "fluorescent lamp ballasts, number C82.2".

2/ Test procedure for performance shall be in accordance with ANSI standard "method of measurement of fluorescent lamp ballast, number C82.1".

3/ Sample lot size for inspection shall consist of five ballast units.

4/ Battery discharge test shall be performed for first article testing. Testing shall include the following:
The storage battery shall be capable of supply and maintaining the maximum total associated load for a period no less than 5 hours, and no less than 87.5% rated battery voltage. To determine if the unit complies with this requirement, the battery is to be tested at 55 ^{o}C with its associated charging equipment as follows:

(a) The battery shall be charged to the maximum rated battery voltage in a period of time not exceed 24 hours.
(b) The charged battery shall then be connected to its maximum rated load and permitted to discharge for 5 hours. At the end of the discharge, the battery voltage shall be no less than 87.5 percent or the rated battery voltage or the illumination no less than 60 percent of initial illumination.
(c) At the end of the discharge period the battery shall be charged for a period of time not to exceed 24 hours.
(d) The battery shall be again discharged at maximum rated load for 3.33 hours, and the battery voltage shall be no less the 87.5 percent of the rated battery voltage or the illumination no less than 60 percent of initial illumination at the end of 3.33 hours.
(e) Following the discharge, the battery again shall be recharged to rated battery voltage in a time period not to exceed 24 hours.
(f) The battery then shall be discharged while connected to maximum rated load until "low voltage cutoff" engages. The minimum time for cutoff shall be no less than 5 hours.
(g) At the end of the discharged time, the battery shall be recharged to rated battery voltage in a time period not to exceed 24 hours.
(h) Following recharging, the battery shall be discharged at rated load for 5 hours. At he end of the discharge, the battery voltage shall be no less than 87.5 percent rated battery voltage or the illumination no less than 60 percent of initial illumination.

Custodians:
Navy - SH
DLA - IS

Preparing activity:
Navy - SH
(Project 6250-N001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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